

ALGORITHMS & DATA STRUCTURES (M)

2025-2026

Prof Alice Miller

Office: S133 (second floor, Lilybank Gardens)

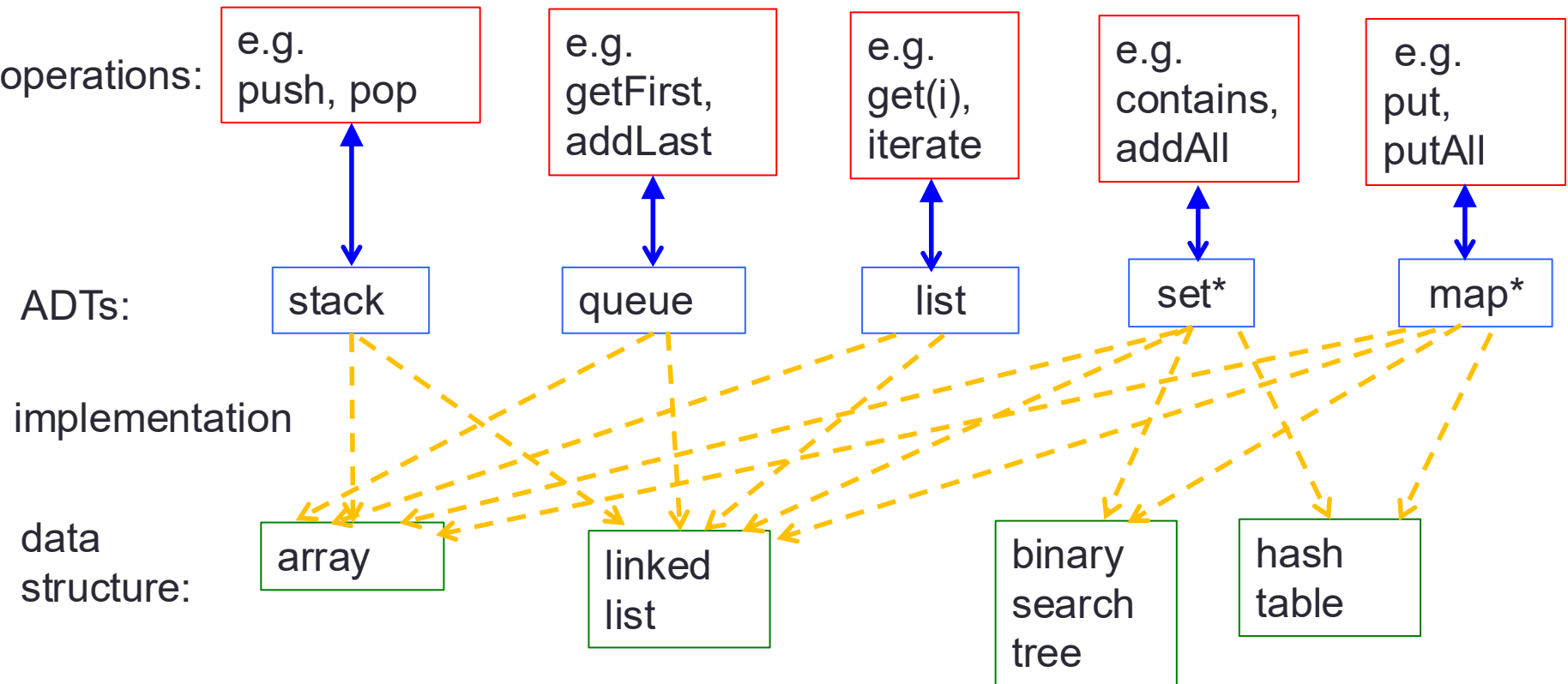
Computing Science → Algorithms & Data Structures (M)

Aims

To study the concepts of:

- **algorithms**
 - and how to analyse their efficiency.
- **abstract data types (ADTs)**
 - the ADTs most commonly used in software development (stacks, queues, lists, sets, maps).
- **data structures**
 - most commonly used to represent these ADTs (arrays, linked-lists, binary-search-trees, hash-tables)
 - + algorithms operating on these data structures.

This is what the course leads to...



It won't mean anything now – but print it out and put it on your wall – we will gradually get there!

Contents (Rough plan)

Topic

- | | |
|---|----------|
| 1. Introduction | Week 1 |
| 2. Algorithms and Complexity | Week 1-2 |
| 3. The Array Data Structure | Week 2-3 |
| 4. Linked-List Data Structures | Week 4 |
| 5. Abstract Data Types (ADTs) | Week 5 |
| 6. Stack ADT | Week 5 |
| 7. Queue ADT | Week 5 |
| 8. List and Iterator ADTs | Week 6 |
| 9. Set ADT | Week 7 |
| 10. The Binary-Search-Tree Data Structure | Week 8 |
| 10A. AVL trees | Week 9 |
| 11. Map ADT | Week 10 |
| 12. Hash-Table Data Structures | Week 10 |

Course operation: synchronous sessions

- One live session with me:
 - Monday 15:00-17:00. Live lecture summaries (i.e. not dwelling on code), Q&A and examples.
- Thurs 15:00-16:00 and 17:00-18:00
 - Labs Labs
- Class rep: ?

Course operation: asynchronous delivery 1

- Lecture slides and videos and lab (tutorial) questions
 - All available via Moodle
 - Material divided into sections (see next slide for example)

4. Linked Lists

Mark as done

In this section we introduce linked lists

Below are links to the powerpoint slides, lecture videos and tutorial questions. There is also a set of revision slides on inner classes [here](#)  .

4. Linked Lists

Slides:

- Slides ([ppt](#)  )

Recorded lectures:

- [4_LLlists_1](#): Introduction to linked lists; introduction to SSL and Node Java classes; printing list and deleting first node (18.10)
- [4_LLlists_2](#): Insertion into a linked list: algorithm, animation and Java implementation (19.07)
- [4_LLlists_3](#): Deletion from a linked list: algorithm, animation and Java implementation (13.44)
- [4_LLlists_4](#): Searching in a singly linked list; doubly linked lists: insertion, deletion and search; comparison of SLL and DLL algorithm complexities (24.33)

Tutorial exercises:

You should attempt some of these questions as before your lab. Your lab tutor will go through any of the questions that you have difficulty with in the lab.

None of these questions are compulsory - they are provided to help with your understanding of the material only.

Solutions are provided in case you get stuck. A list of recommended questions is given but this is only a guide:

Recommended questions to try: 4A, 4B, 4C, 4D, 4E

- [Exercises4.pdf](#)  
- [Solutions4.pdf](#)  

Code from lectures:

- [linkedLists.zip](#)

Course operation: asynchronous delivery 2

- Lecture slides and videos.
 - You should work through these at your own pace. A **suggested schedule** (colour codes) is given on slides 4 and 15, but you may wish to go faster one week and slower the next.
 - I will make material available for several weeks ahead.
- Tutorial questions
 - There are a lot of these (provided for each section)
 - Suggestions given for which to do
 - Will be used in the tutorial/labs, but they are **there for you to look at** when you have been through the videos/slides for each section. **They are not compulsory**. Don't prepare complete solutions for them all, just use them to force you to think about the material (and sketch out solutions). Solutions provided if you get stuck.

Course operation important points:

- The videos include a lot more clarification than the slides themselves, so make sure you watch them the first-time round.
- A 10-credit course demands 100 hours of effort (not including exams). So, you should be spending more than the allotted 3 hours a week going through the material
- Tutorials/labs (see next slide) are very different to ProgIT labs. It is:
 - not a case of “here is a sheet of questions you must do them and sit at a computer writing Java”
 - More a case of “think about how things are working and draw pictures/sketch out some algorithms”
 - Programming/Coding should **not** be done in the lab sessions.

Tutorials/labs

You will have 1 hour per week for lab/tutorial class (Thursdays).

- Although it's referred to as a lab (as it's in the lab), it's more of a tutorial
- Your tutor may go through some of the tutorial questions on the board ASK THEM if you want them to do this
- Will be as interactive as possible (that's up to you!)
- First labs are this Thursday (Thursday 15th)
 - All labs are live – the tutor will roam amongst you and ask you questions about what you are doing. They may also go through some of the solutions on the board or in a small group
- This Thursday, start looking at the questions in the Introduction section. 1A, 1B, 1D, 1F

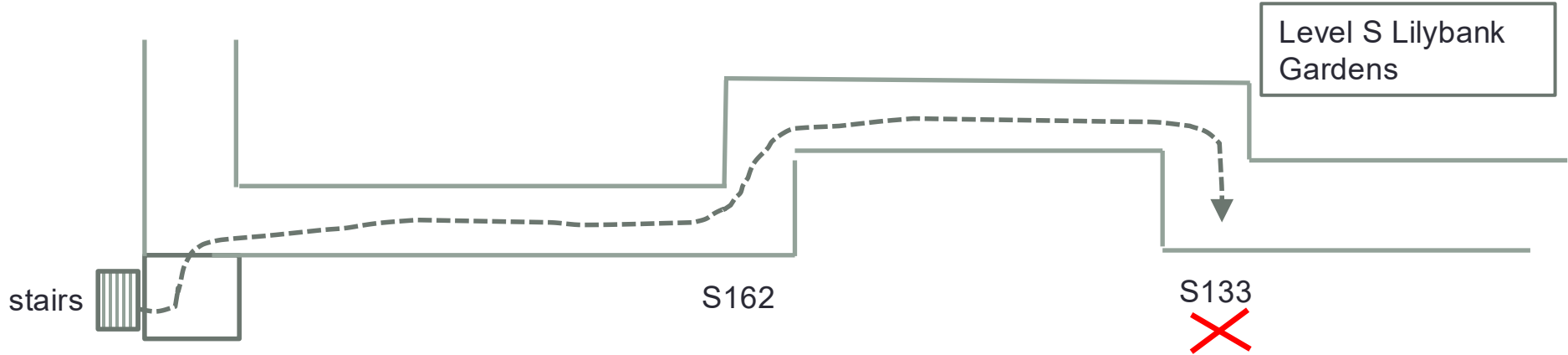
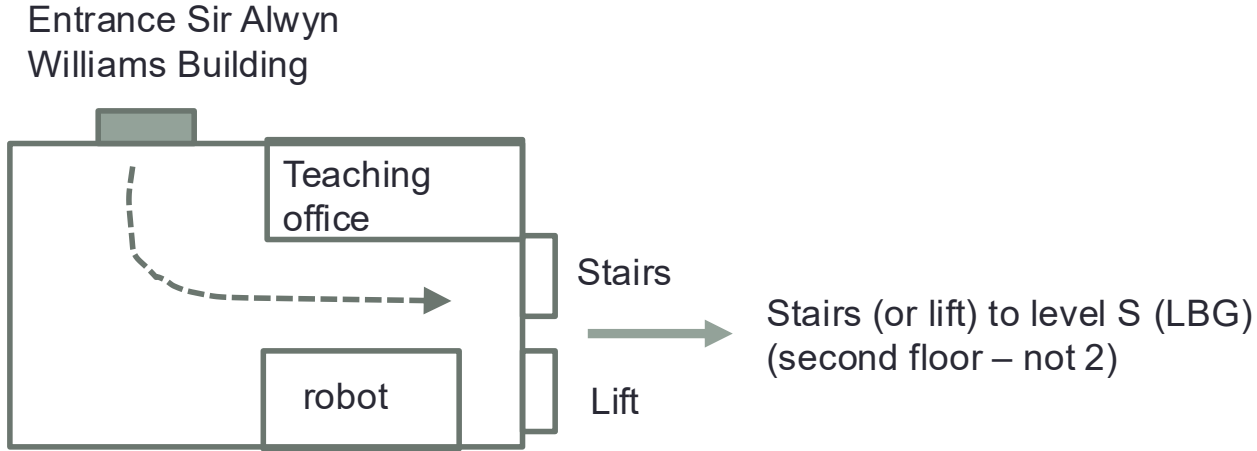
Tutorials/labs continued

- There are two tutors: **Danielius Sukys** and **Frekerik Glitzner**.
- All tutors will be in both lab groups
- I will also come along to the labs and help out if busy
- Don't just sit in lab and work on an assessment for another course
 - Or far away from the rest of the class – or I might not know you are part of the class
 - We will throw people out of the lab if they are not doing ADSIT work and the lab is busy

Office hours

- I will be available Thursday 12:00-13:00 for office hour. Just come along to my office – (S133) or pre-arrange a teams meeting. See map on next slide.
- But I can only guarantee to be free if you let me know that you are coming at a certain time (email me or send me a message via teams). Happy to meet at other times by appointment.
- Meanwhile you can leave questions on the Q&A padlet for me, but **I will only answer when free (I won't be available 24 hours a day!)**. You can also answer questions on there, and I will pick questions from there to address in the Q&A sessions

Map to my office



Assessment

- Assessed Exercise – single exercise, will be handed out **9/3/2025***. Hand in **Mon 23/3/2025** by 16.30 (but it should take you around 6-8 hours to complete)
 - weight = **0.2**
- Examination (this will be computer based but supervised in the lab)
 - weighting **0.8**.
 - Probably quiz/multiple choice (sample exam will be available on Moodle). Will let you know nearer the time. The resit will be non-multiple choice.

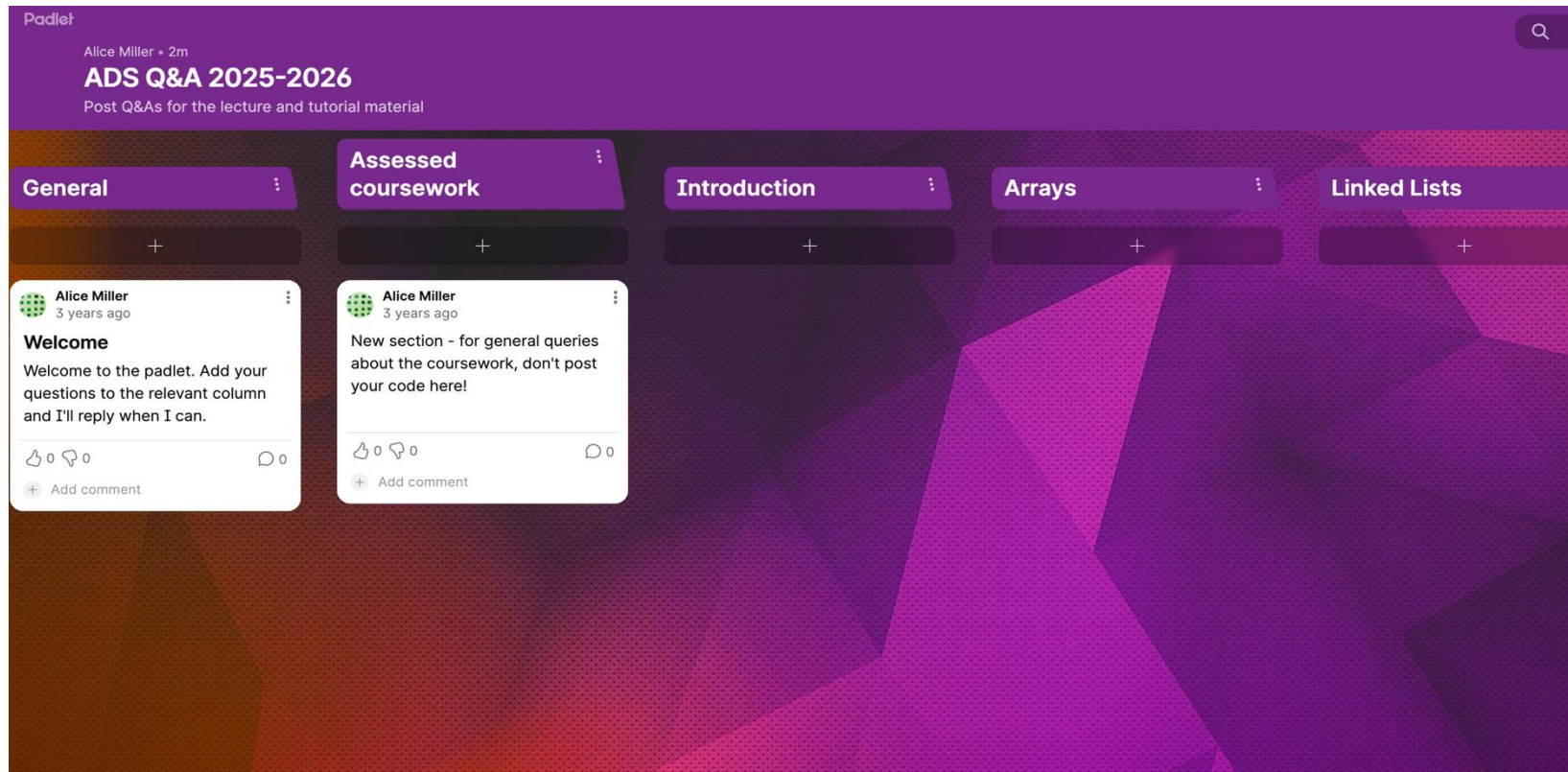
* Subject to change. Waiting on confirmation from the school. Will probably release earlier than that date.

Timetable will be on Moodle (it might change)

Week	Lecture* (MON)	LAB (THURS)	Office hour (Thurs)	Material
1	12 Jan	15 Jan	✓	Introduction + Algorithms and Complexity
2	19 Jan	22 Jan	✓	
3	26 Jan	29 Jan	✓	Arrays
4	2 Feb	5 Feb	✓	Linked Lists
5	9 Feb	12 Feb	✓	ADTs, Stacks, Queues
6	16 Feb	19 Feb	✓	Lists and Iterators
7	23 Feb	26 Feb	✓	Sets
8	2 March	5 March	✓	Binary Search trees
9	9 March	12 March	✓	AVL Trees
10	16 March	19 March	✓	Maps and Hashtables

Q&As via padlet

- For you to ask general questions, or questions about any of the topics
- Ask in the correct column
- If a longer answer or explanation required, I may defer to lecture/Q&A session on the following Friday
- You can answer questions as well as ask them (I'll monitor your answers...). I will respond when I am available, tutors will too
- Keep it polite and respectful! Link is: <https://padlet.com/alicemiller3/ads-q-a-2025-2026-fnbhdf932ndafahv>

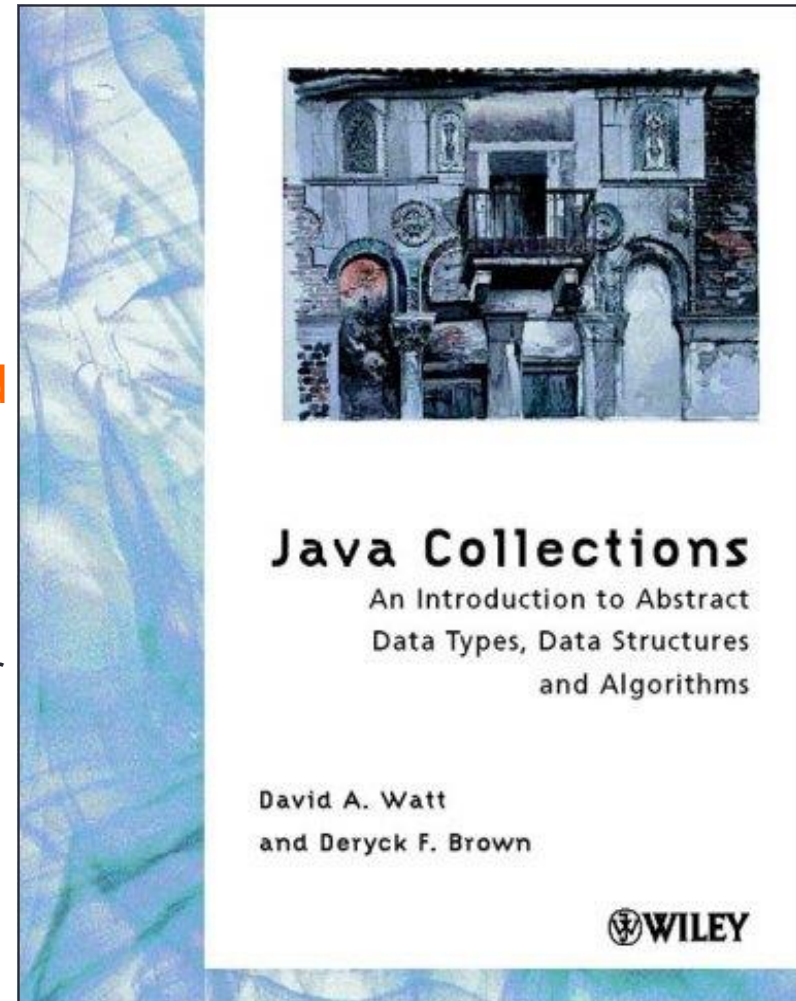


What should I do now?

- After this lecture start going through the **slides and videos**
 - There are also revision slides – (adding, logs, generics) refer to these when you need to (most people will be unfamiliar with **logs** so read this one first if that is you)
 - This week's lab will be on the introduction and the start of the arrays – so take a look at the relevant questions beforehand. The assumption is that you will have also looked at the first part of the arrays slides before the lecture next week (I'll clarify every week at the lecture).
 - Remember the schedule is only a **guide**
- Drop into my **office hour** on **Thursday** if you want a chat (helps if you let me know when you are coming, to make sure I am there).
 - Will be nice to meet you. I can see how you are getting on – and sort out any early problems,
- **This Thursday:** labs, **Next Monday:** Q&A+lecture

Reading (1)

- D.A. Watt & D.F. Brown
Java Collections
Wiley (2002)
 - Useful if you can get hold of a 2nd hand copy but I **wouldn't rush out and buy**
 - Doesn't cover generic classes
 - Some of the later example programs are not "correct" (we'll discuss this later in the course).



Reading (2)

- M.T. Goodrich & R. Tamassia
Data Structures and Algorithms in Java (4th edn)
Wiley (4th edition 2006 or later)
 - optional background reading
(more advanced and mathematical)
- Any algorithms and data structures book using Java from the library
- C. Horstmann, *Big Java* (3rd edn), Wiley (2008)
 - supplementary reading on Java
(covers interfaces, collections, inner classes, generic classes, etc.)
 - You might have this already

Summary of chapters 1&2

- Chapter 1: introduction

- Whistle-stop tour of the concepts of algorithms, data structures and ADTs.
- Don't expect to understand this in detail – but refer back to this lecture throughout the course
- Some example algorithms and algorithm notation (will do algorithms in chapter 2)
- Some example data structures – we will introduce linked lists in chapter 4
- Brief introduction to the idea of ADTs – will do this properly in chapter 5
- Example ADTs (list and set) and how we use Java versions of them – very advanced (chapters 8 and 9) – novice programmers think they understand it when they don't

- Chapter 2: algorithms and complexity

- Introduction to principles of algorithms (some repetition from ProgIT)
- Gentle informal intro to efficiency analysis (how can we compare two algorithms)
- More formal approach
- Lots of examples – refer back to the revision lectures on adding and logs if you get confused
- Big O notation